

From neural SDFs to a node-free boundary inverse

A map of the experiment branches: what each one tried, what it changed in the pipeline, and what survived into today's solver.

26_{/30}

TG-002 cases recovered by the current modal pipeline: 10 shapes × contrasts 0.5 / 4 / 13.3, one centred start (PC-001, ON-001, DP-001, RG-001).

4

cases fail in **every** 30-case campaign, with modal and nodal physics alike: Aphex Twin at all three contrasts and the hook at contrast 13.3.

31_{s vs 157 s}

median time per case, modal vs nodal, at identical recovery (PC-001). A fairer nodal profile takes 101 s (PC-002).

● In pipeline

Part of the default solver today.

● Qualified option

Passed its own test; kept as an option, reference or evidence.

● Negative / closed

Tested and failed its goal, or closed.

● Paused / separate

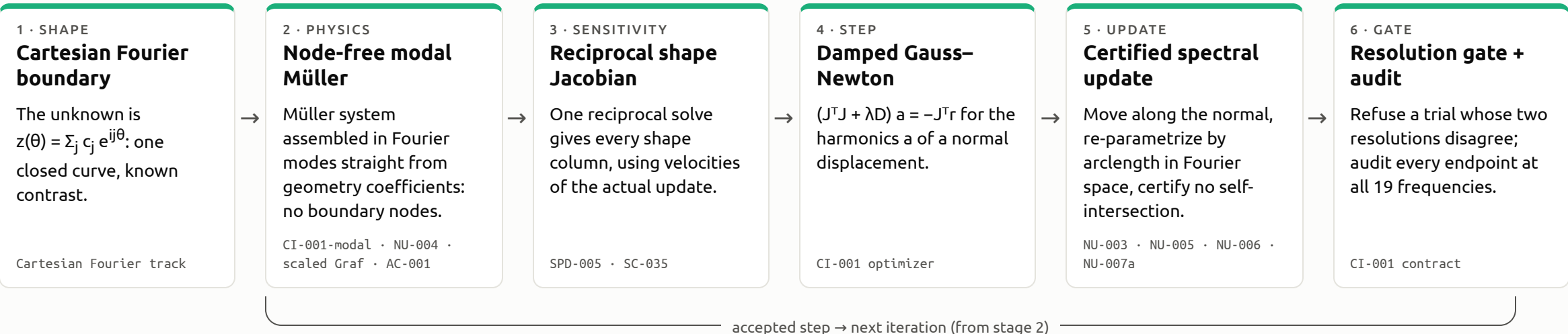
Paused, retired, superseded, or a separate pipeline.

○ Open

Implemented or proposed, not yet run.

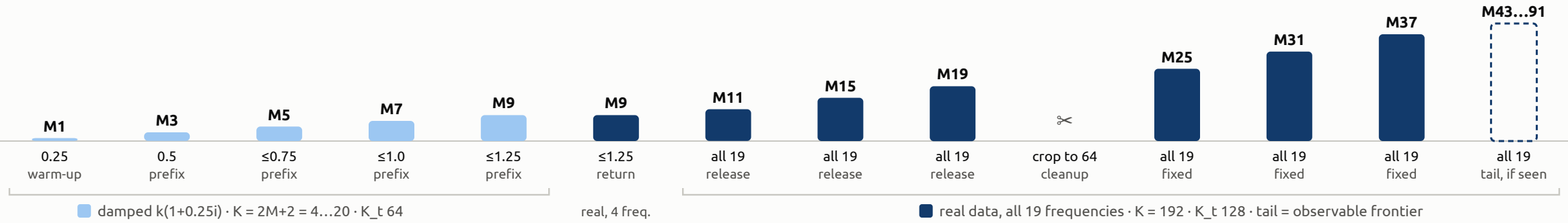
One loop of six stages, wrapped in a continuation schedule

Every stage below came out of a branch on the following slides; the IDs show where.



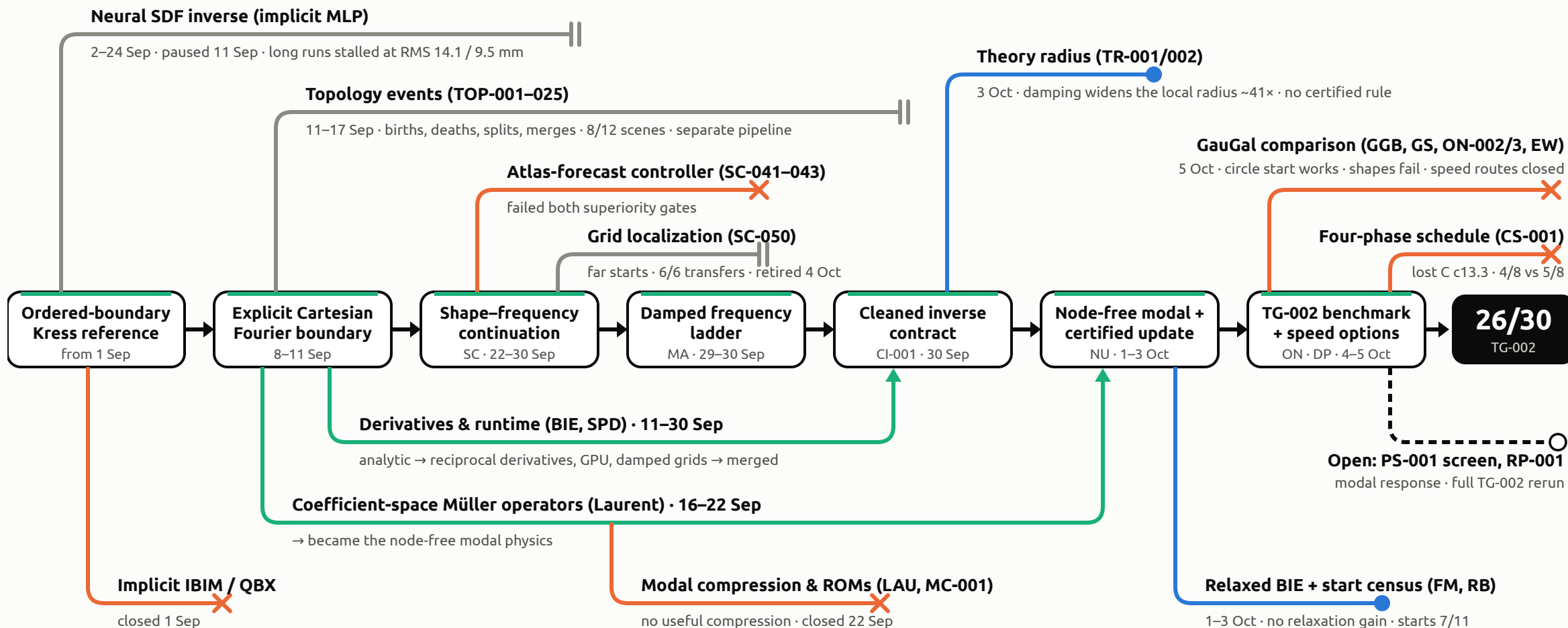
Continuation schedule: damped ladder, then shape release · MA-005 · SC-035 · SC-038 · bar height = shape band M

frequency set (GHz) below each bar; M = normal-update harmonics



Opt-in, not default: ● accuracy exit · ON-001 ● agreement damping + reuse · DP-001 ● decision gate · RG-001 ● four-phase · CS-001 ○ 6 unrun registry candidates · PS-001

Which branches merged into the pipeline, and which ended



Which boundary, and which forward solver?

From an implicit boundary cloud, to ordered nodes, to Fourier coefficients with no boundary nodes in the physics.

Compressed implicit boundary

IBIM · QBX · kdiff – closed 1 Sep

● Negative / closed

CHANGE Solve on a compressed implicit boundary cloud with QBX or kernel-difference quadrature.

RESULT No robustness or cost advantage established; QBX tuning stopped, kept only as a diagnostic.

Ordered-boundary Kress

periodic_kress · gpr_bem_kress

● Reference

CHANGE An ordered closed curve with Kress log-singular quadrature (nodal Nyström).

RESULT The trusted reference: it generated the TG-002 data (N1024/2048-qualified) and is the nodal comparator.

Cartesian Fourier boundary

cartesian_fourier – 10-11 Sep

● In pipeline

CHANGE Optimize the complex Fourier coefficients of $z(\theta)$ directly: no SDF, no star-shape limit.

RESULT Became the state variable of the current inverse.

Node-free modal Müller

CI-001-modal → NU-004

● In pipeline

CHANGE Assemble the Müller system in Fourier modes from geometry coefficients, with no boundary nodes.

RESULT First campaign 19/36 vs nodal 28/36; after fixes, decision-identical to nodal on 6/6 core cases (293 s vs 573 s).

Scaled Graf sources, finer profile

modal fixes – 1 Oct

● In pipeline

CHANGE Rescale the Graf addition series for point sources and receivers; use a finer production resolution profile.

RESULT 7 of the 9 cases that had failed for these reasons now match nodal.

Analytic radial coefficients

AC-001 – 5 Oct

● In pipeline

CHANGE Build the six Chebyshev radial-kernel arrays from Bessel-product formulas instead of sampling.

RESULT 24/24 scalar and 4/4 forward checks pass; slower scalar step (1.00 vs 0.35 ms); TG-002 not yet rerun.

Can an MLP signed-distance function be the unknown?

Paused on 11 Sep by decision: explicit Cartesian Fourier first. The track is scientifically open, not abandoned.

SDF-to-boundary bridge (Method B)

sdf_to_ordered_boundary – from 2 Sep

Qualified option

CHANGE Extract a smooth Cartesian Fourier contour from the SDF zero level set and hand it to Kress.

RESULT The conversion and its reverse derivative work (better than $2e-6$ on audited states). It is a geometry bridge, not an inverse.

Neural SDF inverse through a BEM adjoint

implicit_mlp iterations 1-3 – 8-11 Sep

Negative

CHANGE Optimize MLP weights with the BEM adjoint pulled back through Method B (paired and multistatic data).

RESULT Both long star runs stopped with no decreasing neural step: RMS 14.1 mm (paired), 9.5 mm (multistatic); the multistatic run shrank the 12.5 mm lobe to 1.5 mm.

Iteration-3 repair plan

implicit_mlp/iteration_03/03_plan.md

Open

CHANGE An agreed plan to repair neural geometric motion, written after a recorded gradient diagnosis.

RESULT Never executed. Paused as written, not withdrawn.

Gradient diagnosis

implicit_mlp iteration 3 reviews

Qualified

CHANGE Audit the Kress shape gradient and the Method-B reverse at the stalled states.

RESULT Both cleared (composite within $1e-9$ to $6.4e-6$), so the stall is not a gradient bug. Both runs ended on the admissibility boundary.

The repository and its name still carry this origin, but no neural component runs in the TG-002 pipeline. The explicit radial-Fourier controller built alongside it (8–11 Sep) moved to the topology track.

In what order should frequencies and shape detail be released?

Shape–frequency continuation (SC, 22–30 Sep) and the modal atlas (MA, 29–30 Sep) produced today's schedule.

Centred state band, complete-trial derivative

SC-035

In pipeline

CHANGE

Store the curve at $K = 8/12/16/20$ during the ladder; differentiate the actual update construction.

RESULT

Peanut/C/kite RMS 0.14/0.47/0.57 mm, against 2.94/3.20/2.98 mm; star 16% worse.

All-frequency shape release

SC-038

In pipeline

CHANGE

After the ladder, fit all 19 frequencies while releasing $M = 11, 15, 19$.

RESULT

C 0.49 → 0.024 mm and kite 0.55 → 0.11 mm against matched $M = 9$ controls.

Damped frequency ladder

MA-004 → MA-005

In pipeline

CHANGE

Localize and run the low-frequency prefix at complex wavenumber $k(1 + 0.25i)$.

RESULT

MA-004 repaired 2 of 4 (its gate needed 3). MA-005: 7/8 transfer vs 0/8 frozen (damping alone also 7/8).

Observable-frontier tail

MA-005 → policy

In pipeline

CHANGE

Keep adding 6 shape modes while the paired Jacobian at 2.5 GHz still sees them.

RESULT

A conditional tail, $M = 43...91$, in the default schedule.

Atlas-forecast controller

SC-041 → SC-043

Negative

CHANGE

Choose the next shape band from a sensitivity-atlas forecast.

RESULT

Both superiority gates failed: RMS ratio 1.11 vs a fixed schedule, 1.005 vs a stagnation rule.

Ray steps, wider ladder

SC-036 · SC-037

Negative

CHANGE

Step along rays; store a wider K ladder (8/16/32/64/192).

RESULT

Mixed: the star was restored, but the C got 27.7% worse, failing the 25% gate.

Frequency-only control

SC-051

Negative

CHANGE

Release every shape mode ($M = K = 255$) from the first frequency.

RESULT

0/36 recovered, against 34/36 for the preselected established strategies. Shape release is needed.

Grid localization for far starts

SC-049 → SC-050

Retired

CHANGE

A data-only disk search plus a 0.25 GHz warm-up before the ladder.

RESULT

6/6 far-start transfers recovered; retired with far starts on 4 Oct.

How do we move the boundary safely, without nodes?

CI-001 froze the contract on 30 Sep; the NU series (1–3 Oct) replaced the spline and node-based geometry steps (some sampled checks remain).

Cleaned inverse contract

CI-001 – 30 Sep

● In pipeline

CHANGE

One truth-blind interface with frozen data and gates: $\text{RMS} \leq 1 \text{ mm}$, Hausdorff $\leq 2 \text{ mm}$, residual $\leq 0.3\%$, audit.

RESULT

Nodal 28/36 under the full contract (34/36 on geometry alone).

Coefficient-normal update

NU-001

● Negative

CHANGE

Displace the Fourier coefficients along the normal directly, with no re-parametrization.

RESULT

2/6 and 3/6 cases matched, with drift flags; the nodal trial map was kept.

Spline-free spectral update

NU-003

● In pipeline

CHANGE

Replace the spline resampler with arclength quadrature in Fourier space.

RESULT

Decision-identical on 6/6 core cases; preparation 2.2× slower.

Coefficient validity certificate

NU-005

● In pipeline

CHANGE

Prove there is no self-intersection from $|W|^2$ coefficient bounds, before any sampled test.

RESULT

Decided all 1,131 checks with no sampled fallback; costs +103 s.

Batched GPU geometry preparation

NU-006

● In pipeline

CHANGE

Evaluate every update projection in one float64 GPU batch.

RESULT

Preparation 150 → 3.9 s; wall time 390 → 242 s on the six core cases.

GPU certificates

NU-007 → NU-007a

● In pipeline

CHANGE

Run the validity certificates on the GPU.

RESULT

NU-007 failed its relative-bound gate. NU-007a (a threshold-scaled gate) passed with 39.5% less wall time.

Spline vs spectral replay

GC-001

● Qualified

CHANGE

Replay 279 recorded moves through both geometry updates, with no physics.

RESULT

278/279 reference-qualified, with identical decisions; output-grid aliasing identified as a dominant discrepancy.

Analytic geometry tangent

NF-001

● Option

CHANGE

Differentiate the discrete projection analytically, not by central differences.

RESULT

Selectable as analytic_spectral; the default still uses 10^{-7} m central differences.

Where does the time go, and what can be removed exactly?

Derivative and runtime work (BIE, SPD: 11–30 Sep) carried over; the TG-002 speed options (ON, DP) came on 4–5 Oct.

Analytic operator derivatives

BIE-004

Superseded

CHANGE Differentiate the boundary operators analytically instead of by finite differences.

RESULT 1,008 operator checks pass; derivatives 1.87× faster. Then superseded by reciprocal derivatives.

Reciprocal shape derivatives

SPD-005 → modal Hadamard form

In pipeline

CHANGE One reciprocal solve with receivers as sources yields every shape column.

RESULT A further 1.77–2.38× over operator derivatives (death/split/merge scenes); the idea is used by the modal Jacobian.

Modal instead of nodal physics

PC-001 · PC-002

In pipeline

CHANGE Same schedule and gates; swap only the forward physics.

RESULT Both 26/30. Median 31 s modal vs 157 s nodal; a fairer nodal profile takes 101 s.

Accuracy exit

ON-001 E

Qualified option

CHANGE Stop once the all-frequency residual is $\leq 0.3\%$ and the endpoint audit passes.

RESULT 26/30 kept; median 1.55× faster per successful case. The reach-clip and 7-frequency arms were rejected.

Agreement damping + exact reuse

DP-001 F

Qualified option

CHANGE Set damping from model/actual agreement, skip the terminal tangent, reuse exact solves, batch audits.

RESULT 26/30 kept; all 26 successes faster, median 1.13×.

Stage-entry reuse, validity-first

PS-001 candidates

Open

CHANGE Carry each stage's endpoint solves into the next stage; run the cheap sampled validity test first.

RESULT Implemented as exact (same accepted decisions); no campaign run yet.

Exact geometry reuse

SPD-008

Qualified option

CHANGE Reuse geometry reports and audits while the curve is unchanged.

RESULT 22–55% less time on the topology scenes, with unchanged paths.

Projected modal solve

BIE-002

Negative

CHANGE Solve in a projected, truncated modal basis to save work.

RESULT Reference checks pass, but it runs 3.44× slower; no useful compression.

First-order operator reuse

BIE-006

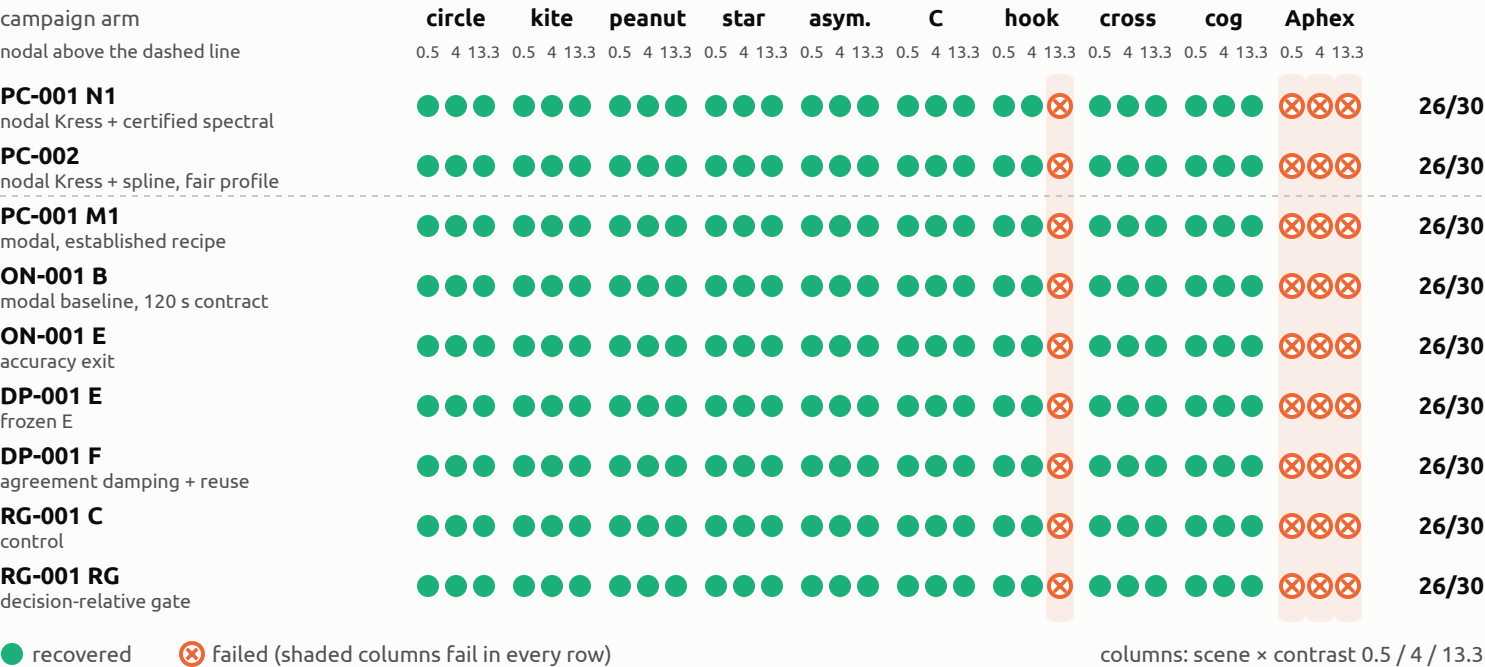
Negative

CHANGE Reuse operators across accepted steps with a tangent correction.

RESULT Most approximations lost accuracy; no qualifying saving. Stopped.

The same four cases fail in every campaign

Each dot is one TG-002 case in one campaign arm, read from its saved result file.



Diagnosis (ON review). Each Aphex run stops on the absolute field-agreement gate, on a trial that still reduced the loss. The hook at c13.3 leaves the damped prefix 6.8 mm off (every success: ≤ 1.13 mm), so it sits in a wrong basin.

Nodal resolution promotion

RB-001 · PC-001 N1

Promote an unresolved trial once to N1024/2048 → still 0/4 (trial wall limits, accuracy-limited trials).

Decision-relative gate

RG-001

Unresolved trials keep their gain decision instead of stopping → 0/4. Aphex c4 RMS 1.69 → 0.66 mm, but its gates fail.

Longer budgets

RG-001 extended

900 s per fit → 0/4; no endpoint passes its audit.

Four-phase schedule

CS-001

Similarity start + four compact all-frequency stages → 3 of the 4 screened, none recovered; it also lost the C at c13.3.

Modal resolution response

RP-001 · modal_response

Promote once to K_t 128/160 and reject there instead of stopping → implemented, not yet run.

Hook basin repair

PX-001

Named in the RG-001 results as possibly needed for the c13.3 basin → not designed.

Ideas that closed or stayed separate

IDEA	IDS	MAIN IDEA / CHANGE	OUTCOME	STATUS
Modal compression & masks	LAU-001-004 · MC-001	Keep only the significant Fourier modes of the operator.	No non-circle 50%-retention mask passed the forward/derivative gate; track closed 22 Sep.	● Negative / closed
Broadband operator ROM	Laurent study, 18 Sep	Represent the operator across frequency with a small basis.	The operator has rank 11 at 10^{-6} , but solutions need about 80; no inverse saving shown.	● Negative / closed
Multi-object local responses	$T = P A^{-1} B$	Couple per-object response operators instead of one big solve.	Matches monolithic solves to 10^{-14} ; not integrated (the benchmark has one object).	● Separate
Topology events	TOP-001-025	Births, deaths, splits and merges of boundary components.	7/12, then 8/12 scenes; a separate pipeline, not promoted to production.	● Separate
Field-defined events	field_defined_events	Topology triggers borrowed from the literature (theta, presence, relocation).	Every borrowed mechanism failed or was blocked.	● Negative / closed
Relaxed BIE, start census	FM-001-005 · RB-001	A relaxed boundary-integral formulation; a 512-start census for the hard C (legacy CI-001 cases).	Relaxation added no recovery; selected endpoints 4/11 → 7/11 with resolution promotion.	● Bounded
Theory radius	TR-001 · TR-002	Measure the local convergence radius; test a certified handoff rule.	Damping widens the empirical radius about 41×; the handoff rule passed 0 of 7 applicable cases.	● Empirical only
GauGal comparison	GGB-001-005 · GS-001	Run against the GauGal pixel solver; add exact translation/radius first stages.	Exact translation/radius stages find the circle centre to 0.1–0.2 mm, but shapes fail; GauGal's own fit reaches 0.46% residual at only IoU 0.33 (GS-001).	● Negative
GauGal speed routes	ON-002 · ON-003 · EW-001	A GauGal adapter, and an Ewald-split Green's function on a Fourier grid.	Adapter 0/25 solve gates; the Ewald circle control failed, with a 46–48× contraction-cost bound.	● Negative / closed
Exploratory extensions	exploration, 2 Oct	Fresnel data, TE/lossy media, half-space, 3-D IBIM sphere, time domain.	Feasibility studies only; not part of any TG-002 claim.	● Separate

What we can claim, and what is already queued

Supported by the evidence

- A maintained 2-D, single-boundary, known-contrast inverse: Cartesian Fourier geometry, node-free modal Müller physics, reciprocal sensitivities, a certified spectral update and a damped shape–frequency schedule.
- On the noiseless TG-002 benchmark it recovers **26/30**, the same count as nodal physics, at about one-fifth of the median time.
- Accuracy exit and agreement damping + reuse keep all 26 and run faster (median 1.55× and 1.13×).
- The four failures are shared by every physics and update tried. The Aphex runs end on the resolution gate, and relaxing that gate alone did not recover them; the hook sits in a wrong basin.

Open threads already in the record

- **Policy screen** PS-001: 13 registered recipes, committed in 71087669; the referenced plan file is missing and no screen has run.
- **Modal resolution response** RP-001 · `modal_response`: implemented, unqualified.
- **Full TG-002 rerun** on the current code, since the AC-001 kernel change has had only component tests.
- **Hook basin** PX-001 and **blind start transfer** FM-006: proposed only.
- **GauGal hybrid** GP-001: proposed, awaiting approval.

Not supported yet: a global convergence guarantee, robustness to noisy data, arbitrary topology recovery, end-to-end sample-free certification, recovery by neural-SDF optimization, or superiority over GauGal.